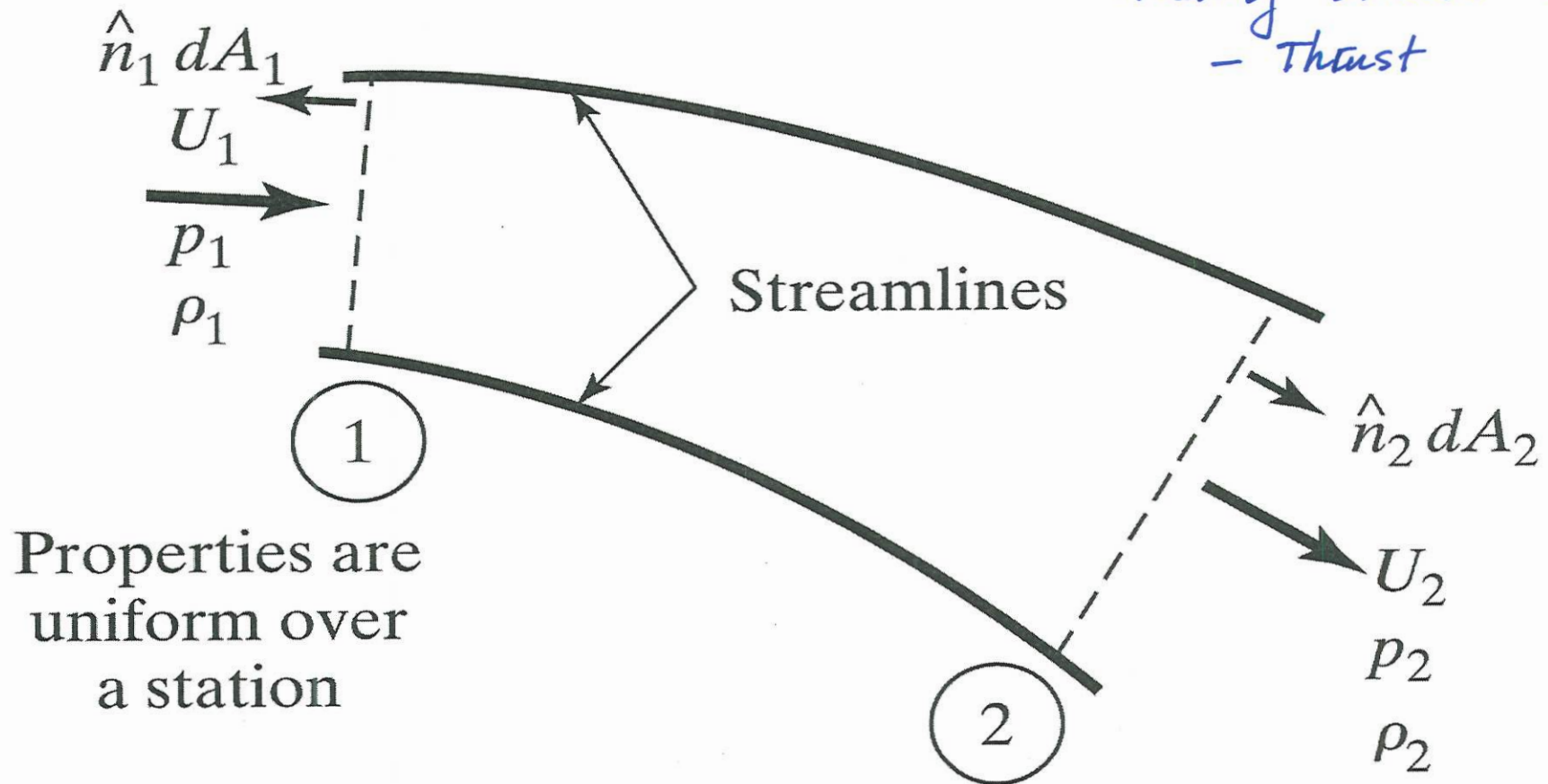


Compressible Flow Through Nozzles [8.3, 8.4] + notes

Objective: Study flows in converging diverging nozzles

*including shock waves
- Thrust*



Conservation of mass: $\rho_1 U_1 A_1 = \rho_2 U_2 A_2$

$$\rho U A = \text{constant} \Rightarrow d(\rho U A) = 0$$

$$(d\rho)UA + \rho(dU)A + \rho U(dA) = 0$$

$$\Rightarrow \frac{d\rho}{\rho} + \frac{dU}{U} + \frac{dA}{A} = 0 \quad (8.49)$$

Euler's equation (3.5) $d\left(\frac{U^2}{2}\right) + \frac{dp}{\rho} = 0$

Rewrite and use (8.26) $UdU = -\frac{dp}{\rho} = -\frac{dp}{d\rho} \cdot \frac{d\rho}{\rho} = -a^2 \frac{d\rho}{\rho}$

$$\Rightarrow \frac{d\rho}{\rho} = -\frac{UdU}{a^2} = -M^2 \frac{dU}{U}$$

$$\left(-M^2 \frac{dU}{U} + \frac{dU}{U} + \frac{dA}{A} = 0 \Rightarrow \right)$$

put back to (8.49) $(1-M^2) \frac{dU}{U} + \frac{dA}{A} = 0$

$$\frac{dU}{U} = -\frac{dA/A}{(1-M^2)} \quad (8.50)$$

1) $0 \leq M < 1$ (subsonic flow)

for $dU > 0 \Rightarrow dA < 0$

for $dU < 0 \Rightarrow dA > 0$



To increase velocity you need a convergent duct

same trends as $\rho = \text{ct}$ flow

Note: for $M = 0 \Rightarrow \frac{dA}{A} = -\frac{dU}{U} \Rightarrow AU = \text{constant}$

2) For $M > 1$ (supersonic flow)

for $dU > 0 \Rightarrow dA > 0$

for $dU < 0 \Rightarrow dA < 0$



diverging



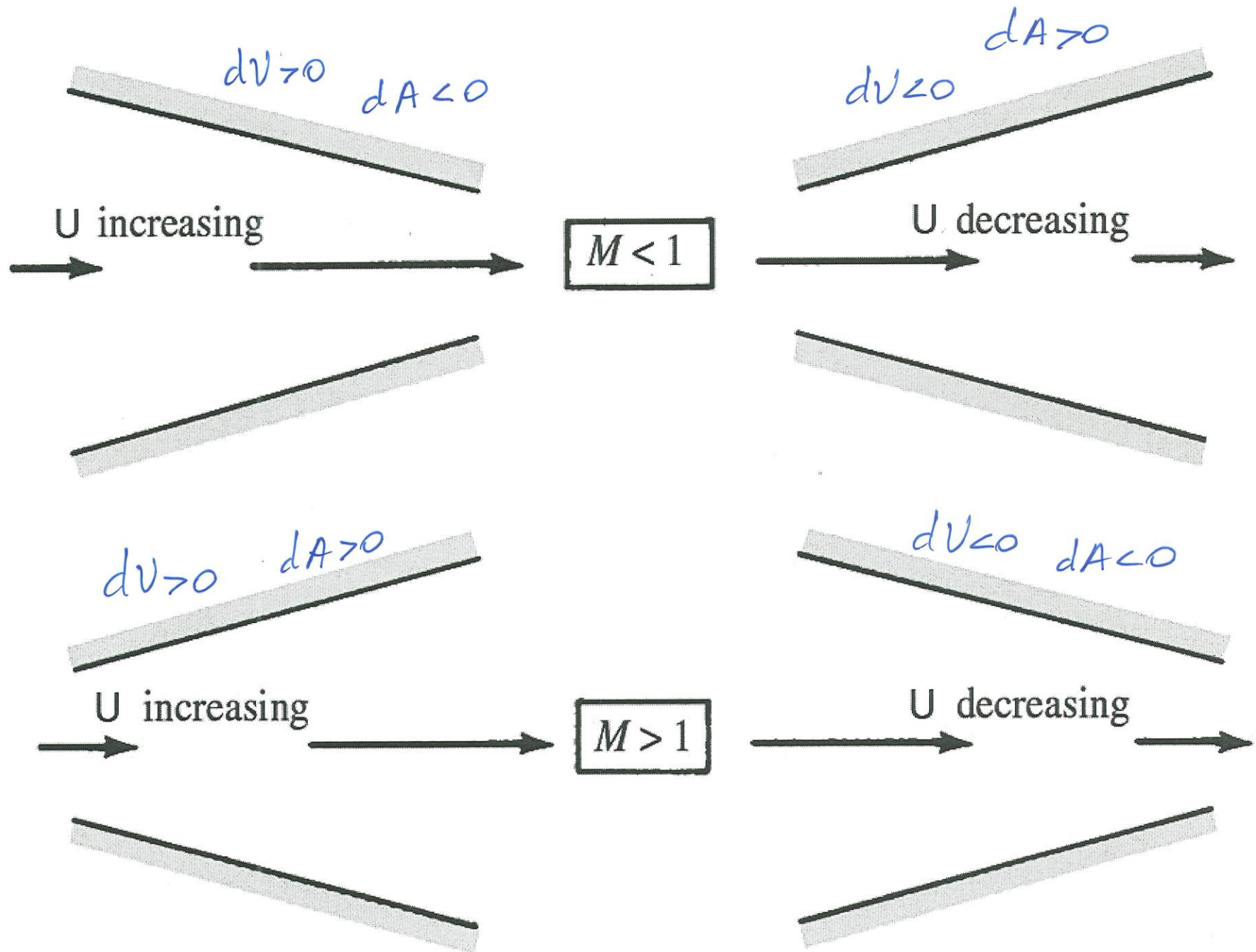
converging

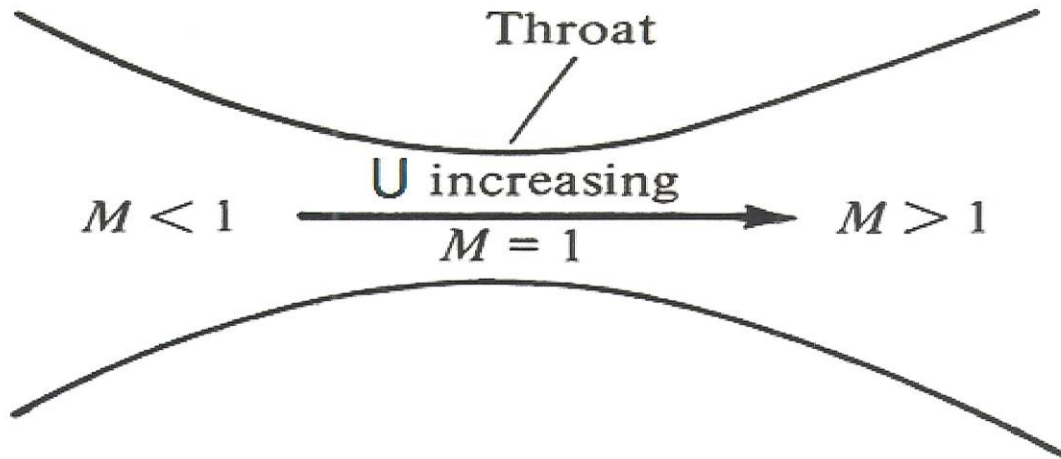
To increase velocity you need a divergent duct (opposite to $M < 1$)

3) $M = 1$ (sonic flow)

$$dA = 0 \quad \text{for} \quad dU \neq 0 \quad \Rightarrow \quad \text{max or min in } A(x)$$

Actually it is a min, convergent-divergent duct

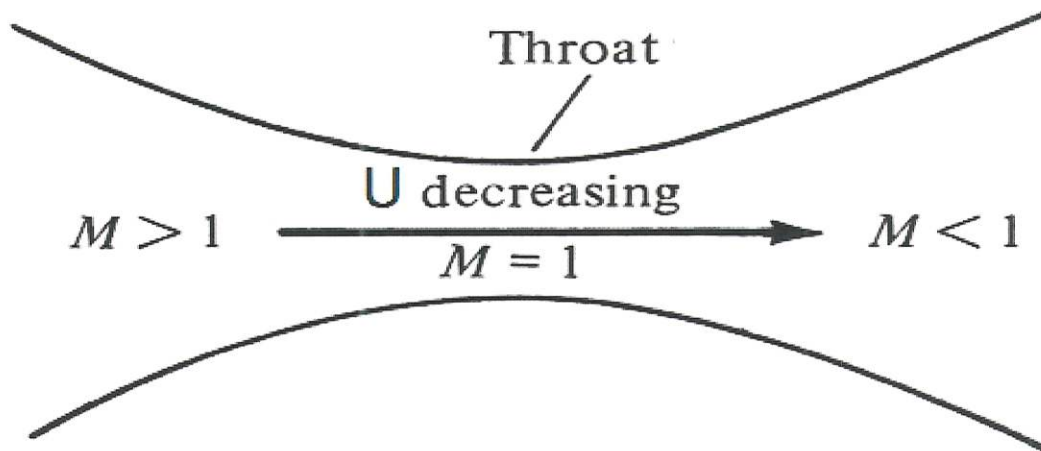




converging-diverging nozzle

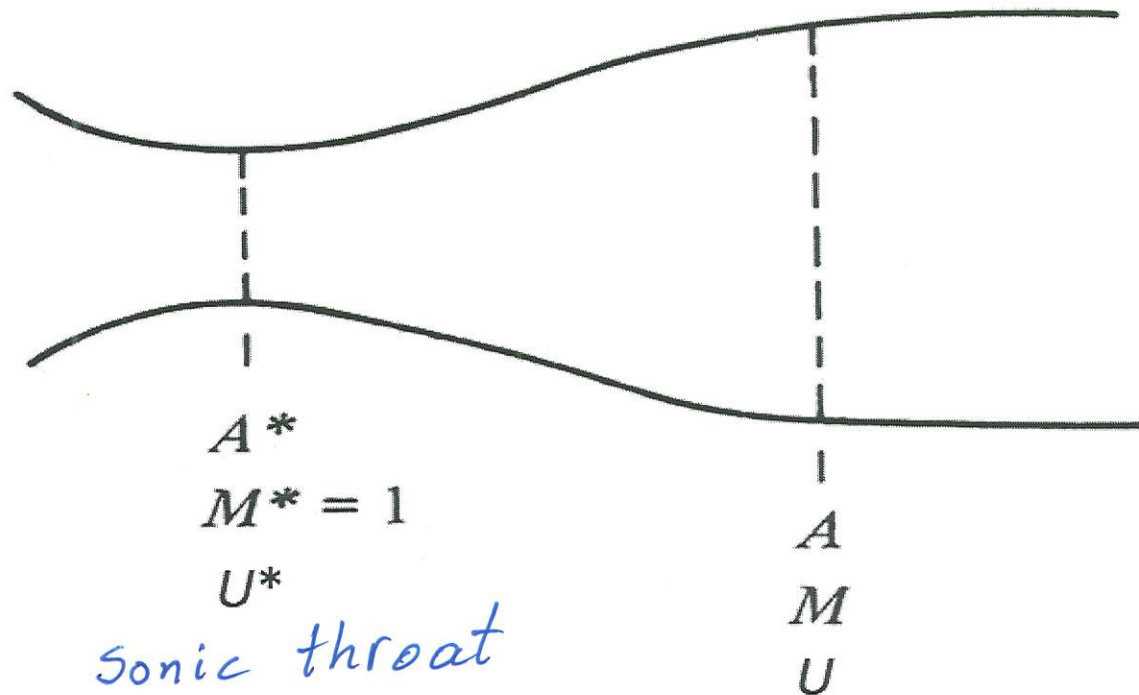
Supersonic
nozzle

only way to achieve
supersonic flow
(e.g. rockets)



Supersonic
diffuser

only way to decelerate
supersonic flow to
subsonic



* denotes sonic conditions

continuity: $\rho^* U^* A^* = \rho U A \Rightarrow \frac{A^*}{A} = \frac{\rho U}{\rho^* U^*} = \frac{\rho_t (\rho/\rho_t) \cdot \sqrt{\gamma R T_t} \sqrt{T/T_t} M}{\rho_t (\rho^*/\rho_t) \sqrt{\gamma R T^*} \sqrt{T^*/T_t}}$

use isentropic relations for * ($M=1$)

$$\Rightarrow \left(\frac{A^*}{A} \right)^2 = \frac{1}{M^2} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{-(\gamma + 1)/(\gamma - 1)} \quad (8.43)$$

Area - Mach number relation (part of isentropic tables)

inverse: $M = f(A^* / A)$

given in isentropic tables e.g. $\frac{A^*}{A} = 0.5 \Rightarrow \begin{matrix} M = 0.31 \\ M = 2.2 \end{matrix}$

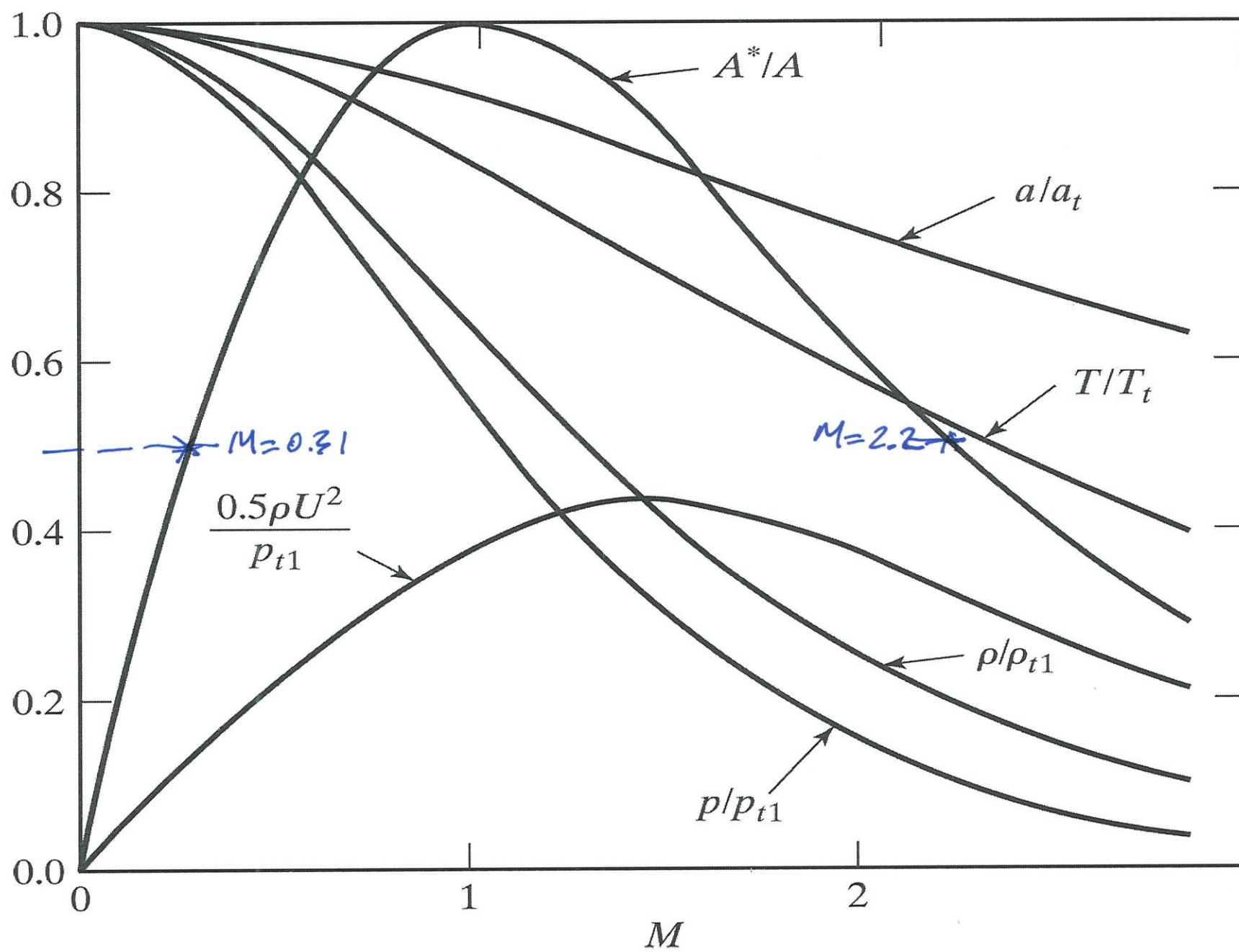
$A^*/A < 1 \Rightarrow 2$ solutions for M

- 1) $M < 1$ (subsonic)
- 2) $M > 1$ (supersonic)

*depends
on back
pressure*

For $M < 1$ as $M \uparrow \Rightarrow A/A^*$ decreases $\downarrow$ till $M = 1$

For $M > 1$ as $M \uparrow \Rightarrow A/A^*$ increases $\uparrow$ (see graph)



use (i) in case there is a shock ($p_{t2} < p_{t1}$)

$$\dot{m} = \rho U A = \rho_{t1} \left(\frac{\rho}{\rho_{t1}} \right) M \sqrt{\gamma R T_t} \sqrt{\frac{T}{T_t}} A$$

Using (8.34) and (8.37) and ideal gas law:

$$\frac{\dot{m}}{A} = \sqrt{\frac{\gamma}{R}} \frac{p_{t1}}{\sqrt{T_t}} \frac{M}{\left\{ 1 + [(\gamma - 1) / 2] M^2 \right\}^{(\gamma+1)/2(\gamma-1)}} \quad (8.44)$$

Max $\dot{m}$ for $M = 1$ ($A = A^*$)

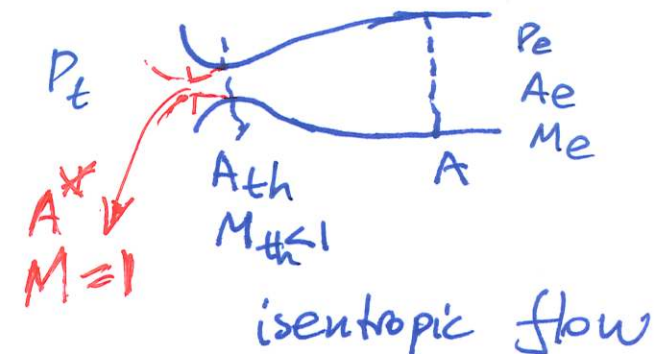
$$\frac{\dot{m}}{A^*} = \frac{p_{t1}}{\sqrt{T_t}} \sqrt{\frac{\gamma}{R} \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/(\gamma-1)}} \quad (8.45)$$

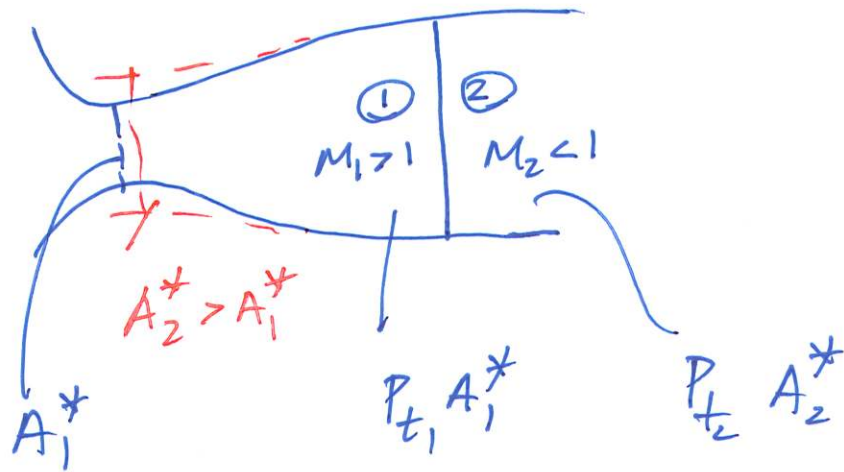
(8.45) valid for not choked flow, too, if we choose the appropriate $A^* \neq A_{th}$ ("equivalent" throat where $M_{th} = 1$)

Also, across a shock $p_{t1} A_1^* = p_{t2} A_2^*$

$$\frac{p_e}{p_t} \Rightarrow M_e \sim \frac{A_e}{A^*} \sim A^*$$

$$\frac{A_{th}}{A^*} \Rightarrow M_{th} < 1$$



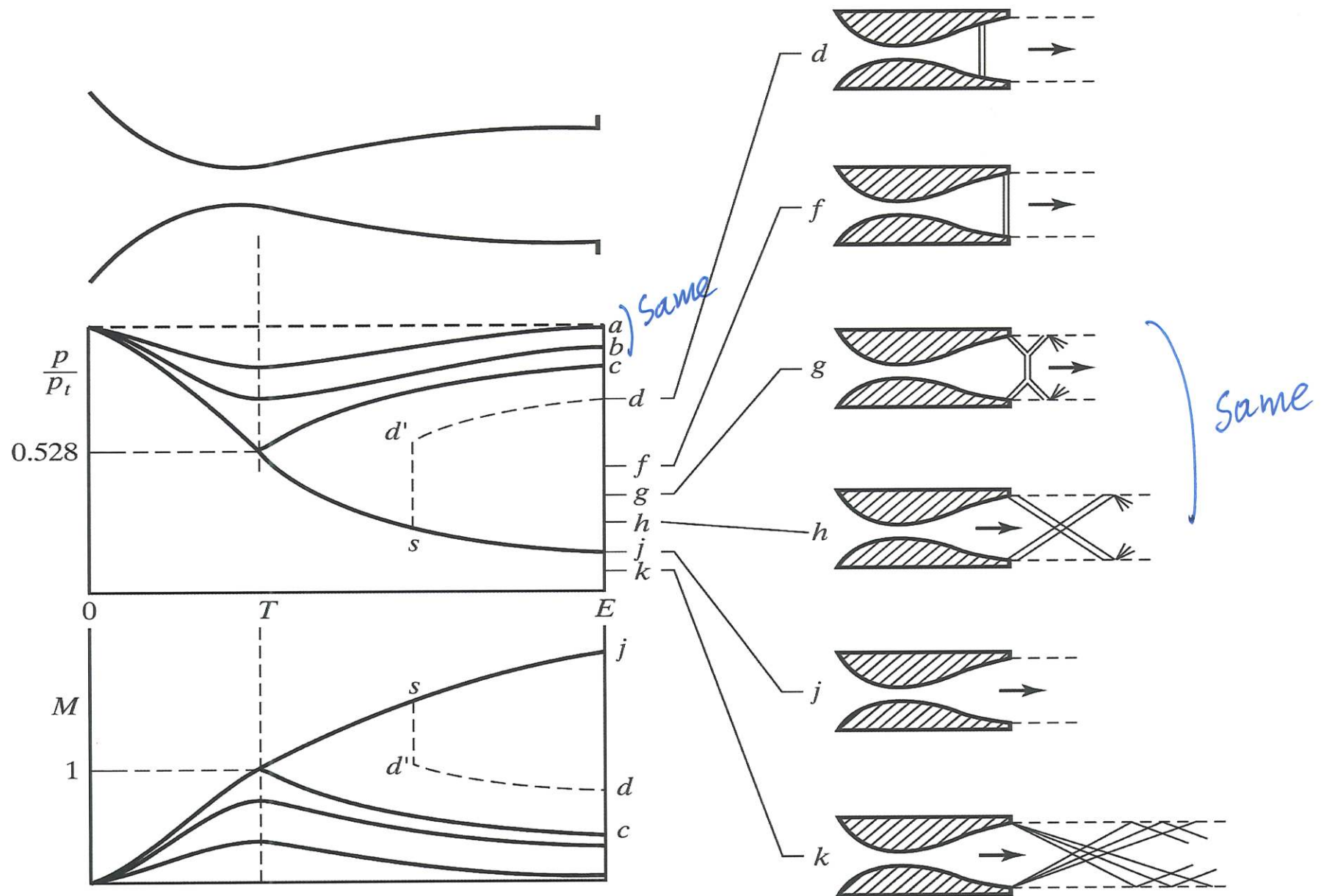


$$P_{t1} A_1^* = P_{t2} A_2^*$$

(mass flow rate is the same across the shock)

$$P_{t2} < P_{t1} \Rightarrow A_1^* < A_2^*$$

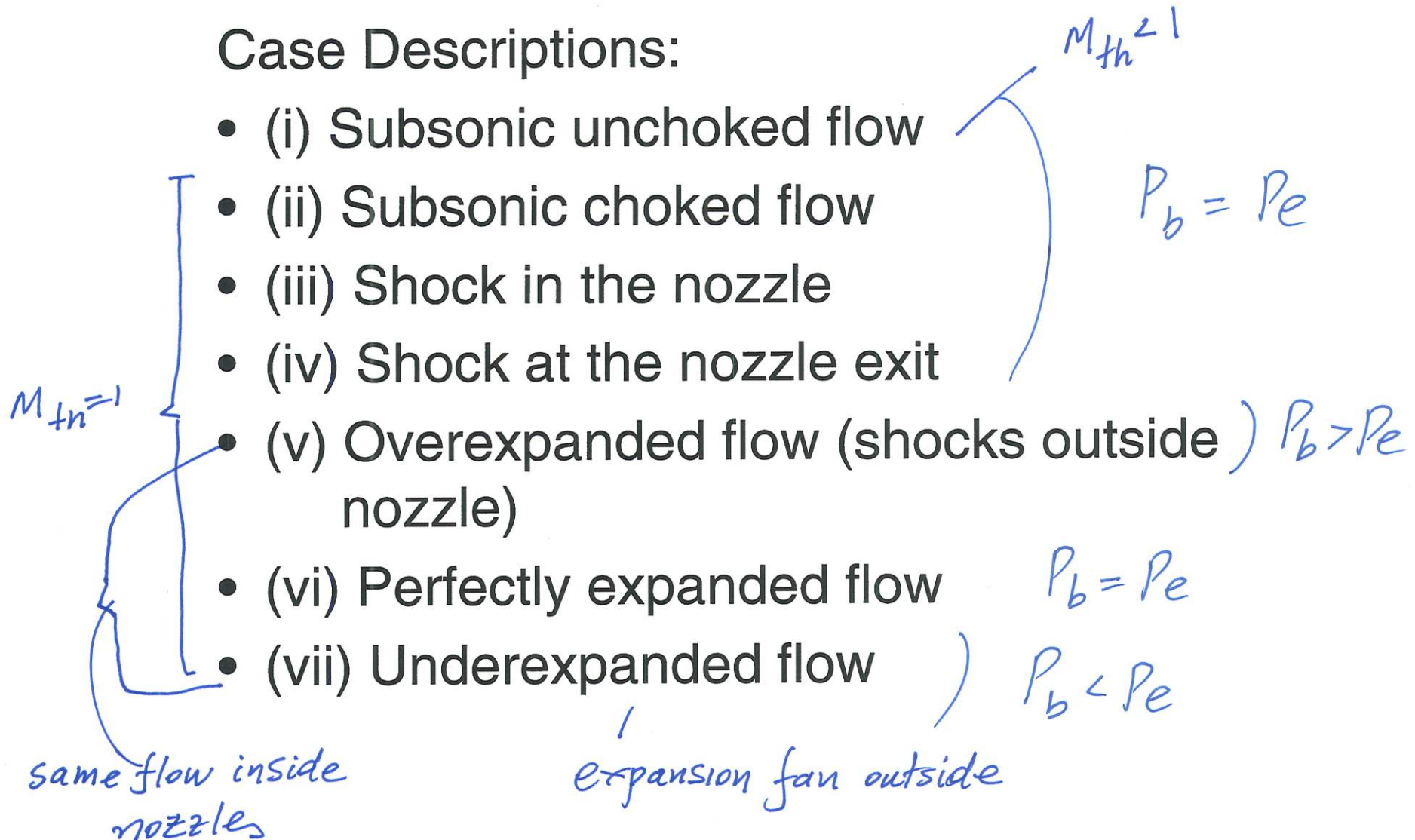
= equivalent throat area after the shock if the flow was isentropic

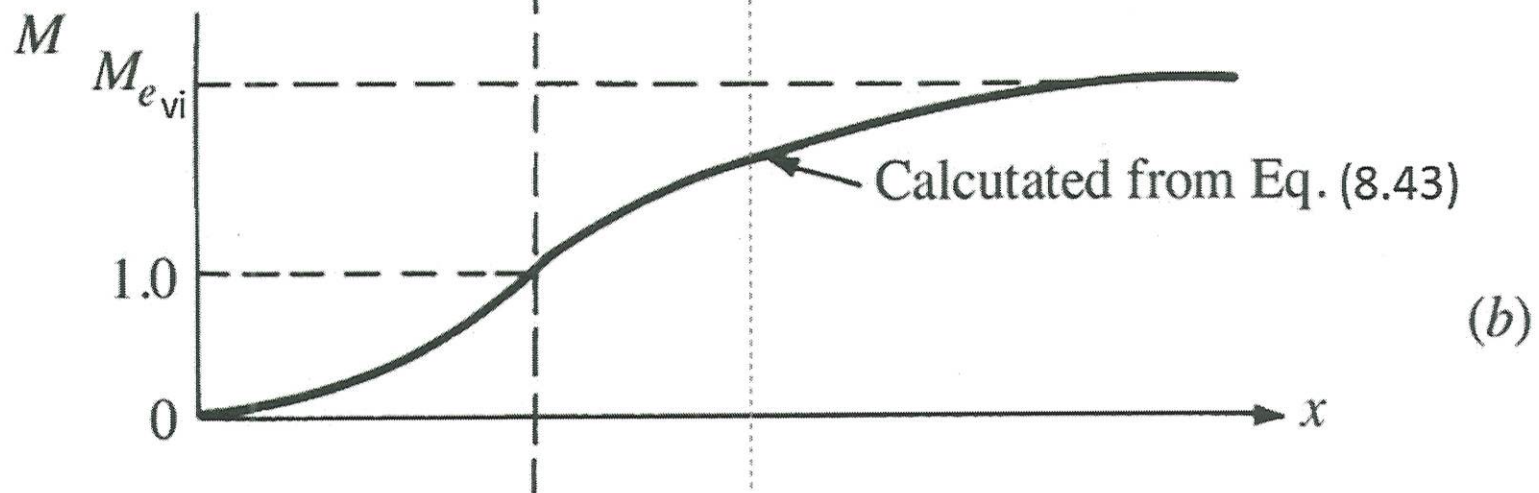
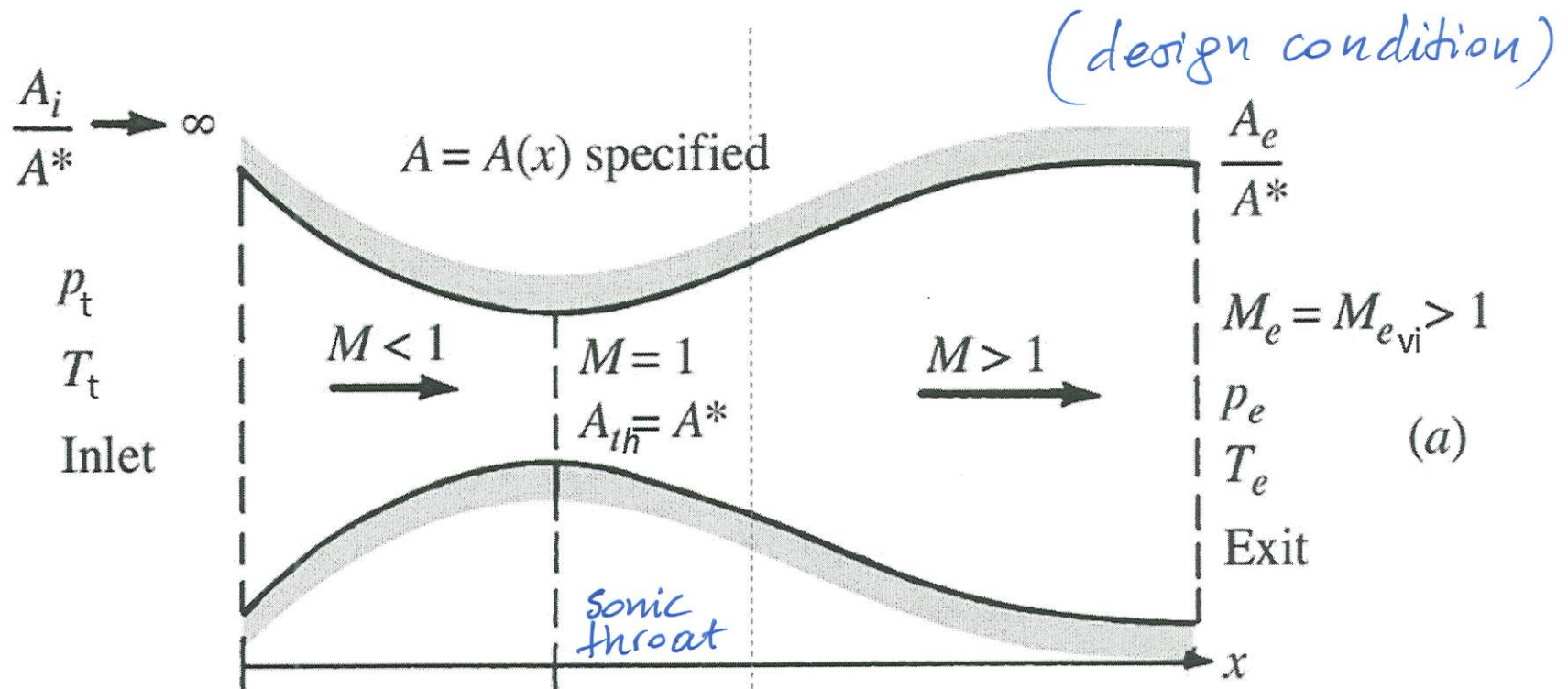


Nozzle Flow Analysis

Case Descriptions:

- (i) Subsonic unchoked flow
- (ii) Subsonic choked flow
- (iii) Shock in the nozzle
- (iv) Shock at the nozzle exit
- (v) Overexpanded flow (shocks outside nozzle)
- (vi) Perfectly expanded flow
- (vii) Underexpanded flow





We need $p_e < p_t$ For the above flow $\frac{p_e}{p_t} = \frac{p_{e\,vi}}{p_t}$

(numbering in page 13)

Review the various cases:

0) $p_e = p_t \Rightarrow$ no flow

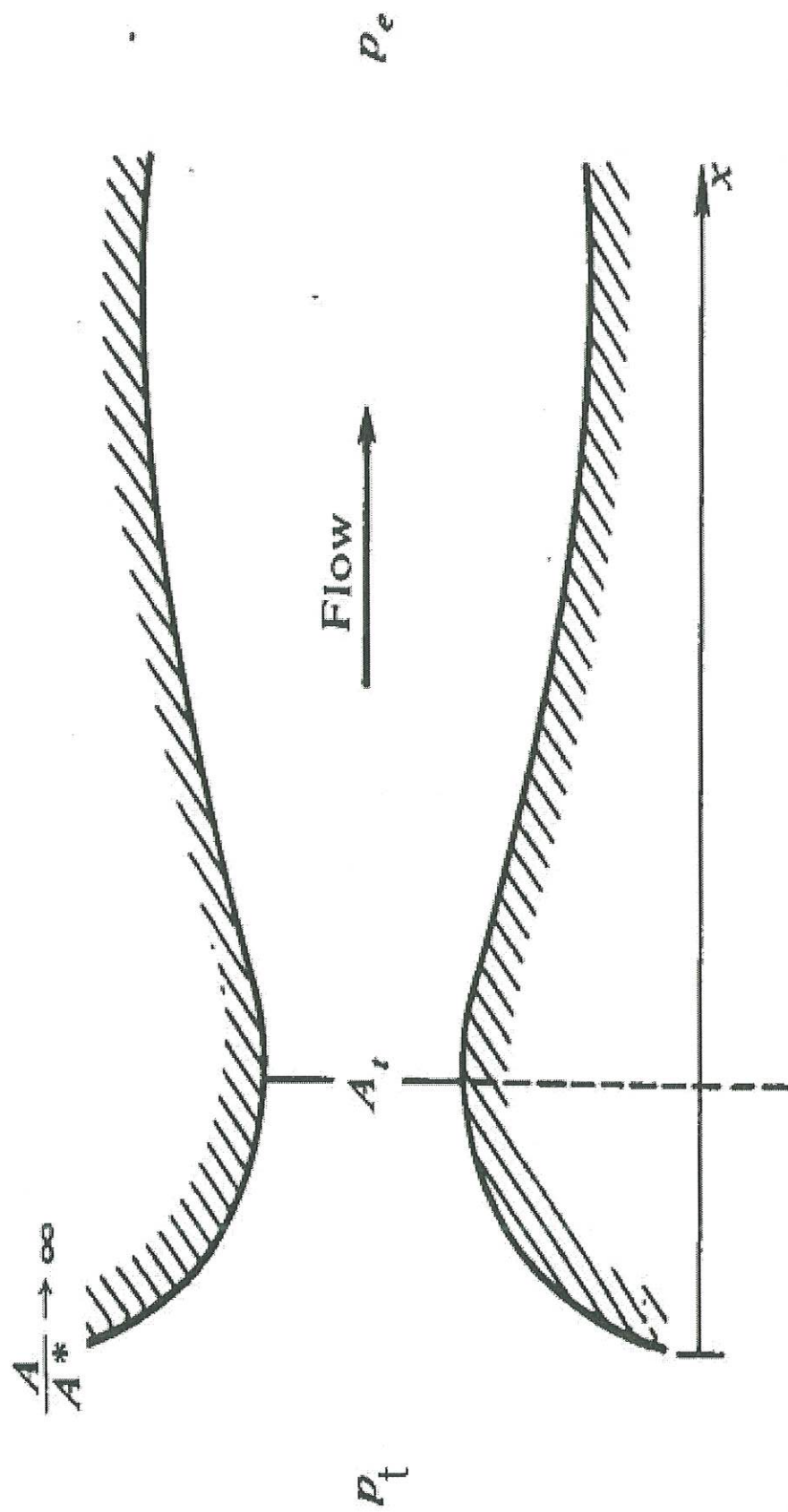
- i) decrease of $p_{e_i} < p_t$ subsonic flow everywhere $M_t < 1$ (unchecked)
- ii) further decrease $p_{e_{ii}} < p_{e_i}$ $M_t = 1$, $M < 1$ elsewhere (checked)

For a given nozzle ($A(x)$) there is only one subsonic – supersonic

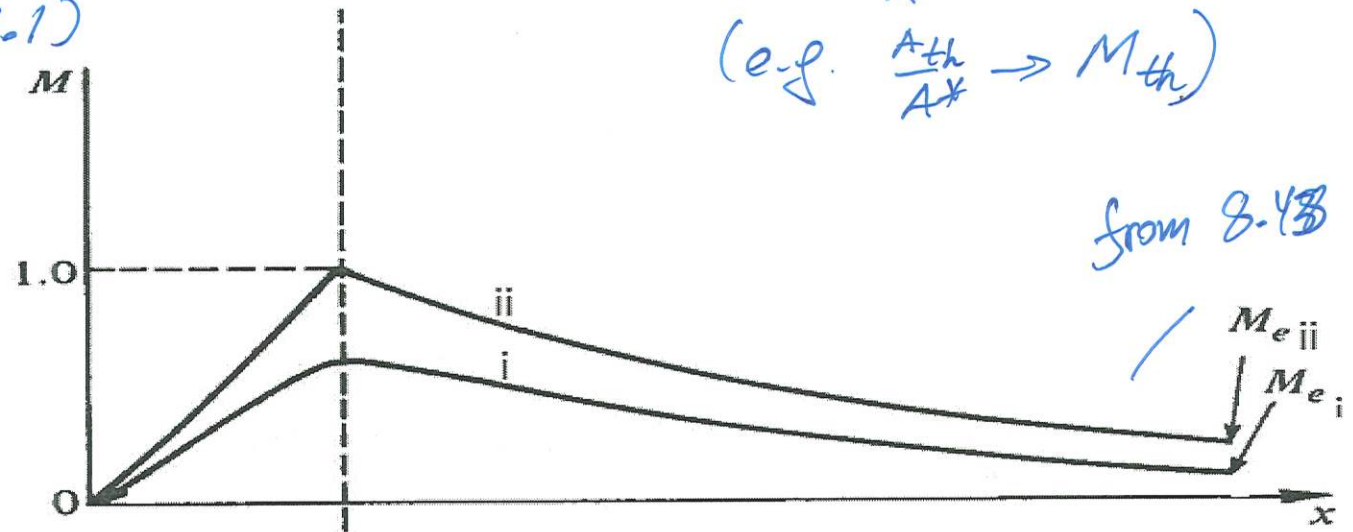
(vi)

isentropic solution but infinite subsonic solutions

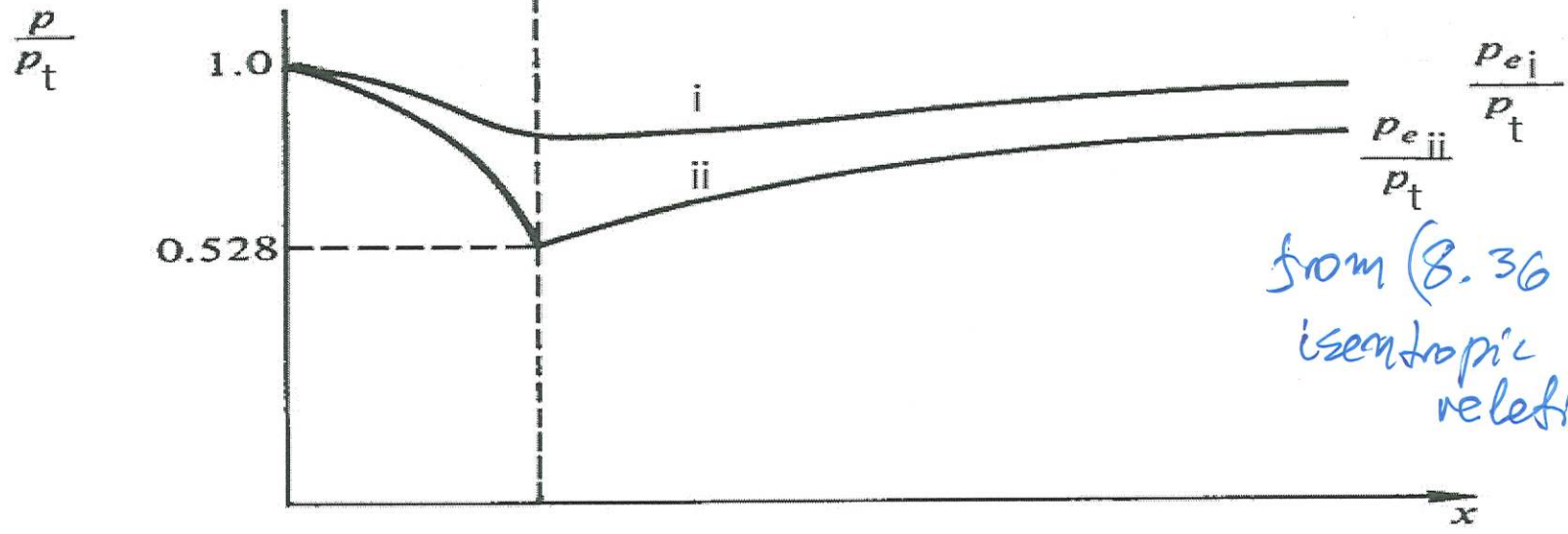
(i)



(i) $\frac{p_e}{p_t} \rightsquigarrow M_e \rightsquigarrow \frac{A_e}{A^*} \rightsquigarrow A^* < A_{th}$ from $\frac{A}{A^*} \rightsquigarrow M$
(table 8.1)
(e.g. $\frac{A_{th}}{A^*} \rightarrow M_{th}$)



from 8.43 $\frac{A^*}{A} = f(M) \rightarrow A = \dots$



from (8.36)
isentropic
relation

same for p, T

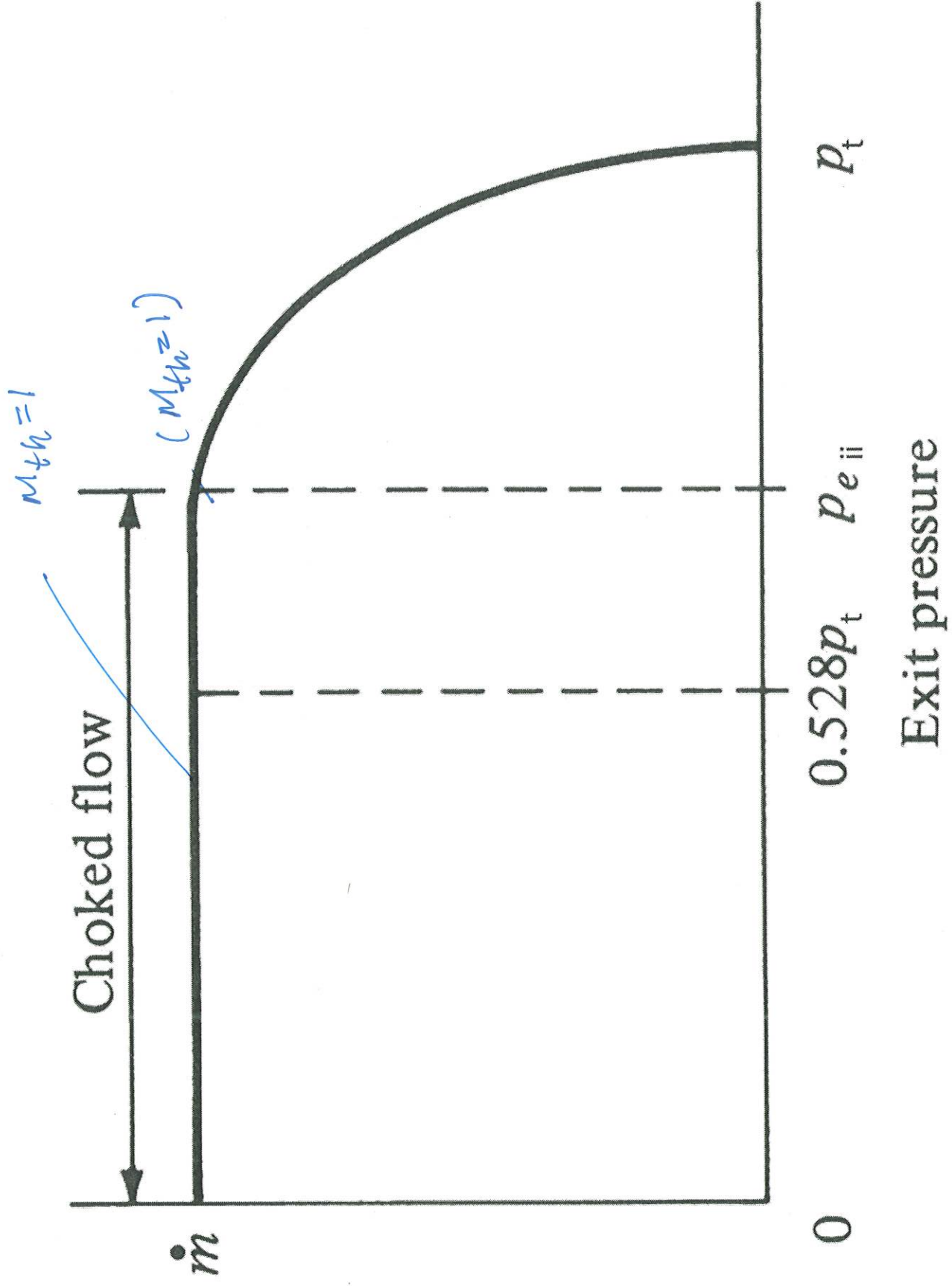
Consider $\dot{m} = \overset{\text{throat}}{\rho_{th} U_{th} A_{th}}$ as $p_e \downarrow$ then $U_{th} \uparrow$ $\rho_{th} \downarrow$

but the increase in U_{th} bigger $\Rightarrow \dot{m} \uparrow$

when $p_e = p_{eii} \Rightarrow \rho U A = \rho^* U^* A^*$

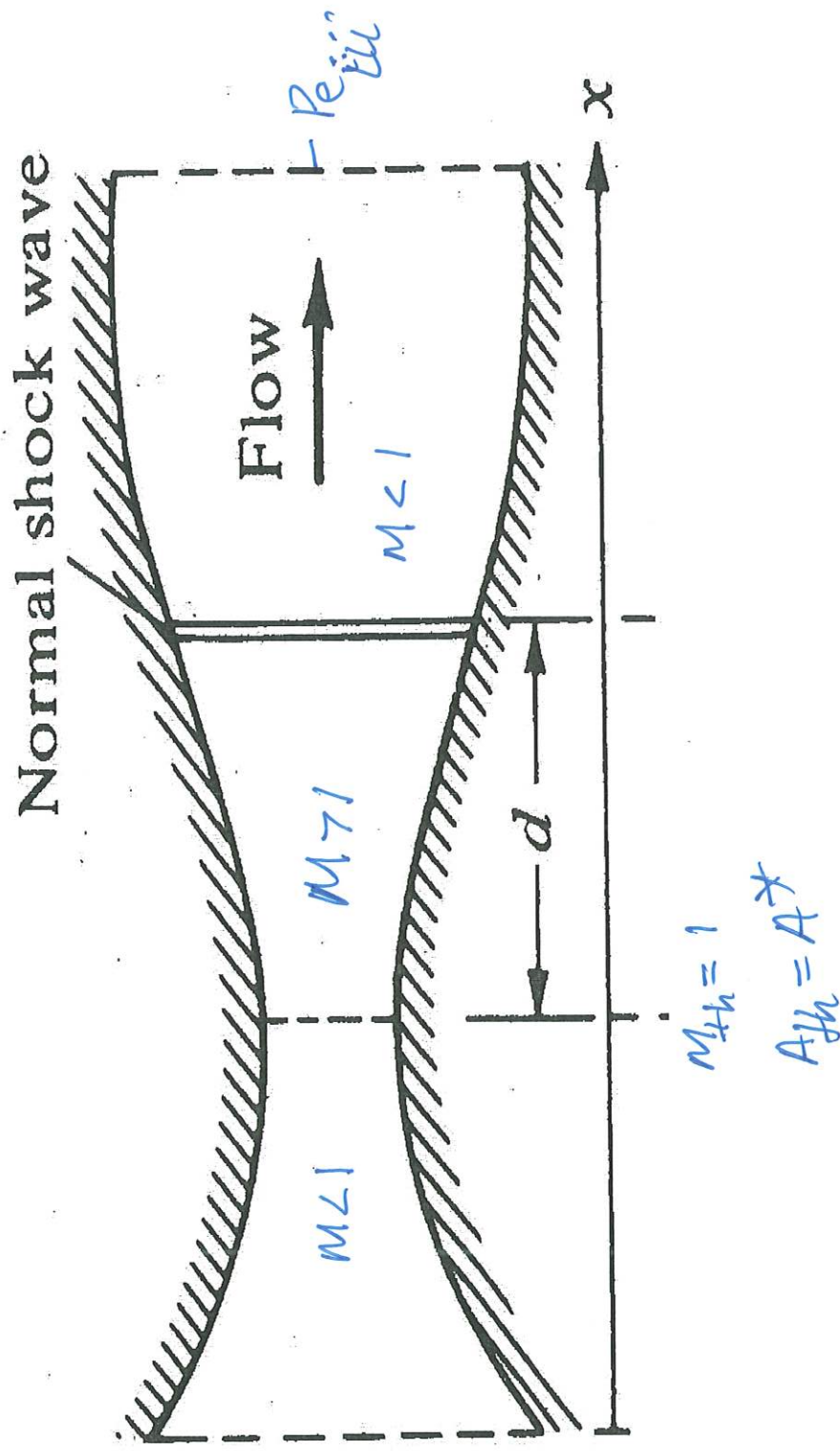
if p_e is further reduced $\Rightarrow \dot{m}$ is unchanged: choked flow

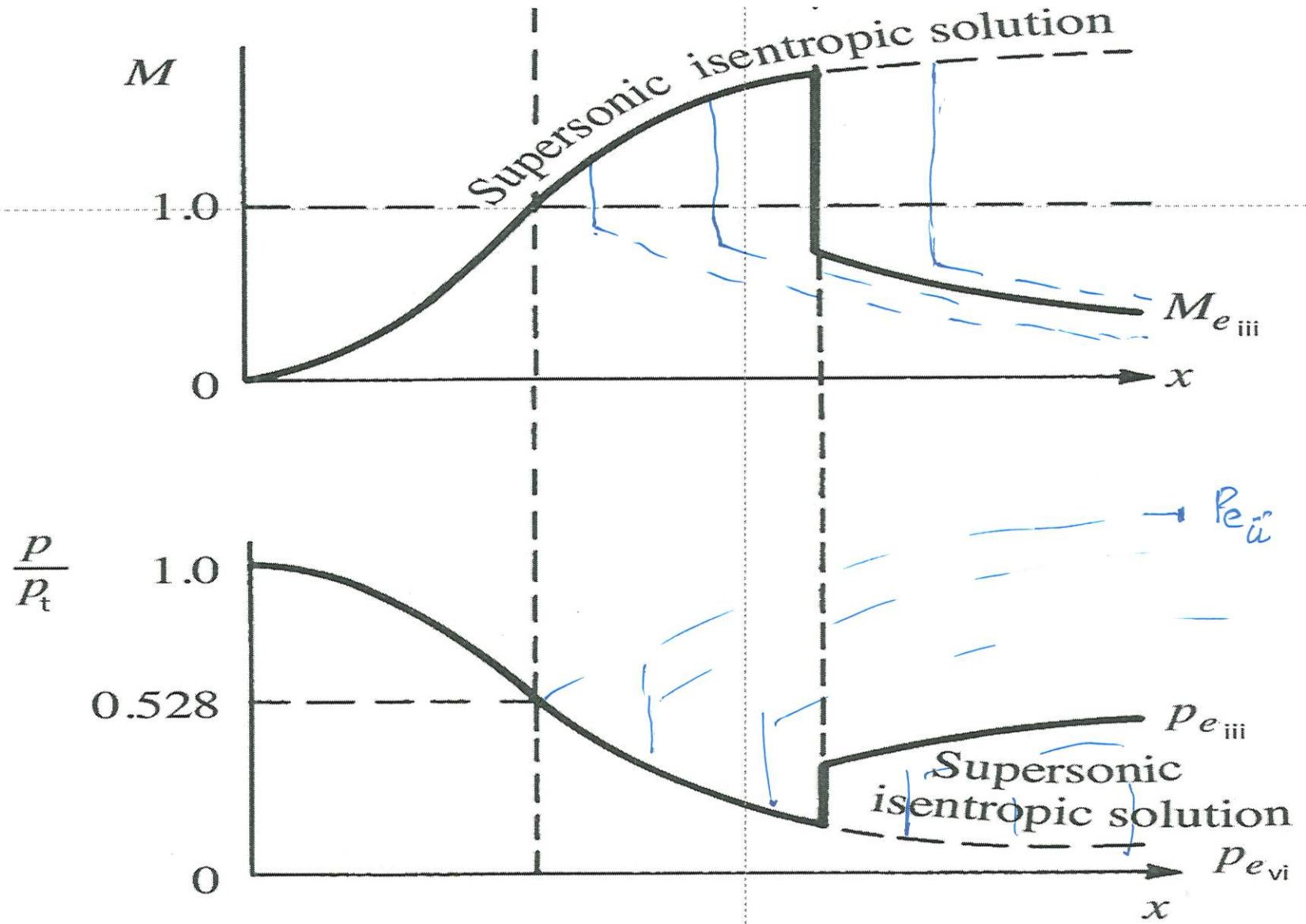
(disturbances cannot work upstream of the throat)



iii) reduce P_e below what happens? ($P_{e_{iii}} < P_{e_{ii}}$)

convergent section: nothing happens; divergent section: shock





behind the shock: isentropic flow

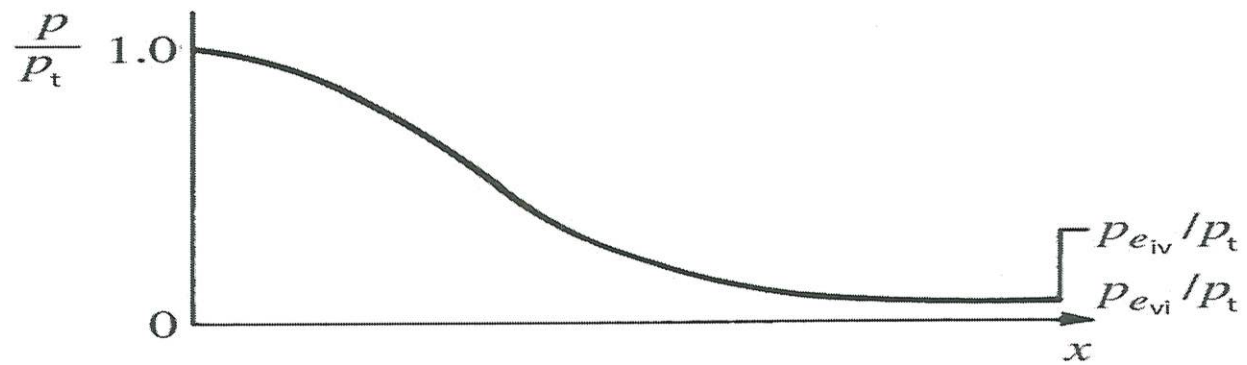
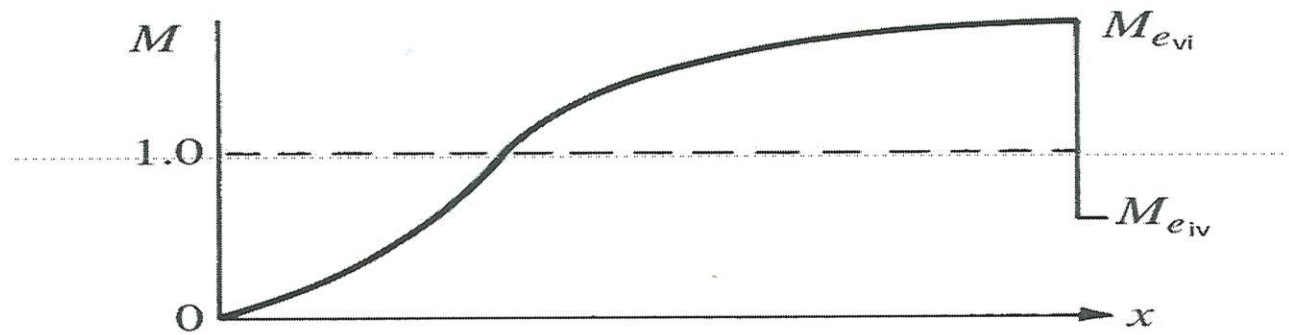
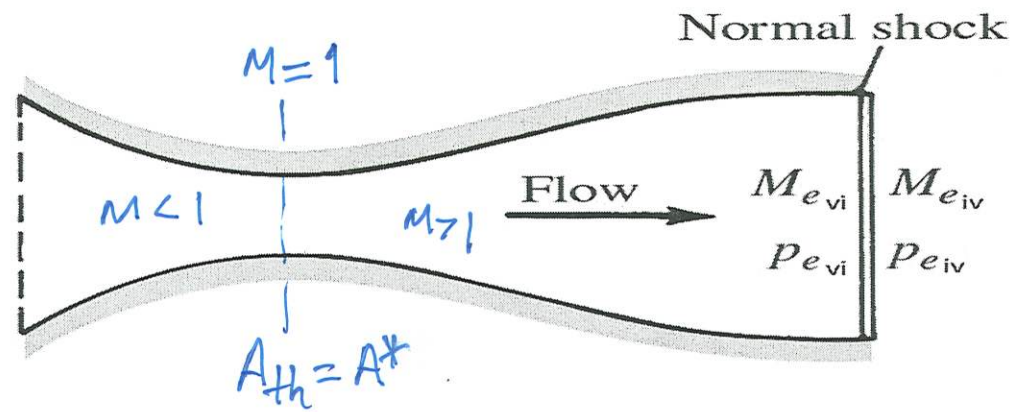
entropy increase across shock ($s_2 > s_1$)

shock location is determined by exit pressure $p_{e_{iii}}$ (see later)

note that across the shock: $p_{t1} A_1^* = p_{t2} A_2^*$

$$(p_{t1} > p_{t2} \Rightarrow A_1^* < A_2^*)$$

iv) as p_e decreases to $p_{e_{iv}}$ the shock moves at the exit



Up to now $p_b = p_e$ This is not always the case

v) For back pressure $p_{e_{iv}} > p_b > p_{e_{vi}}$ there is an oblique shock at the exit – overexpanded jet

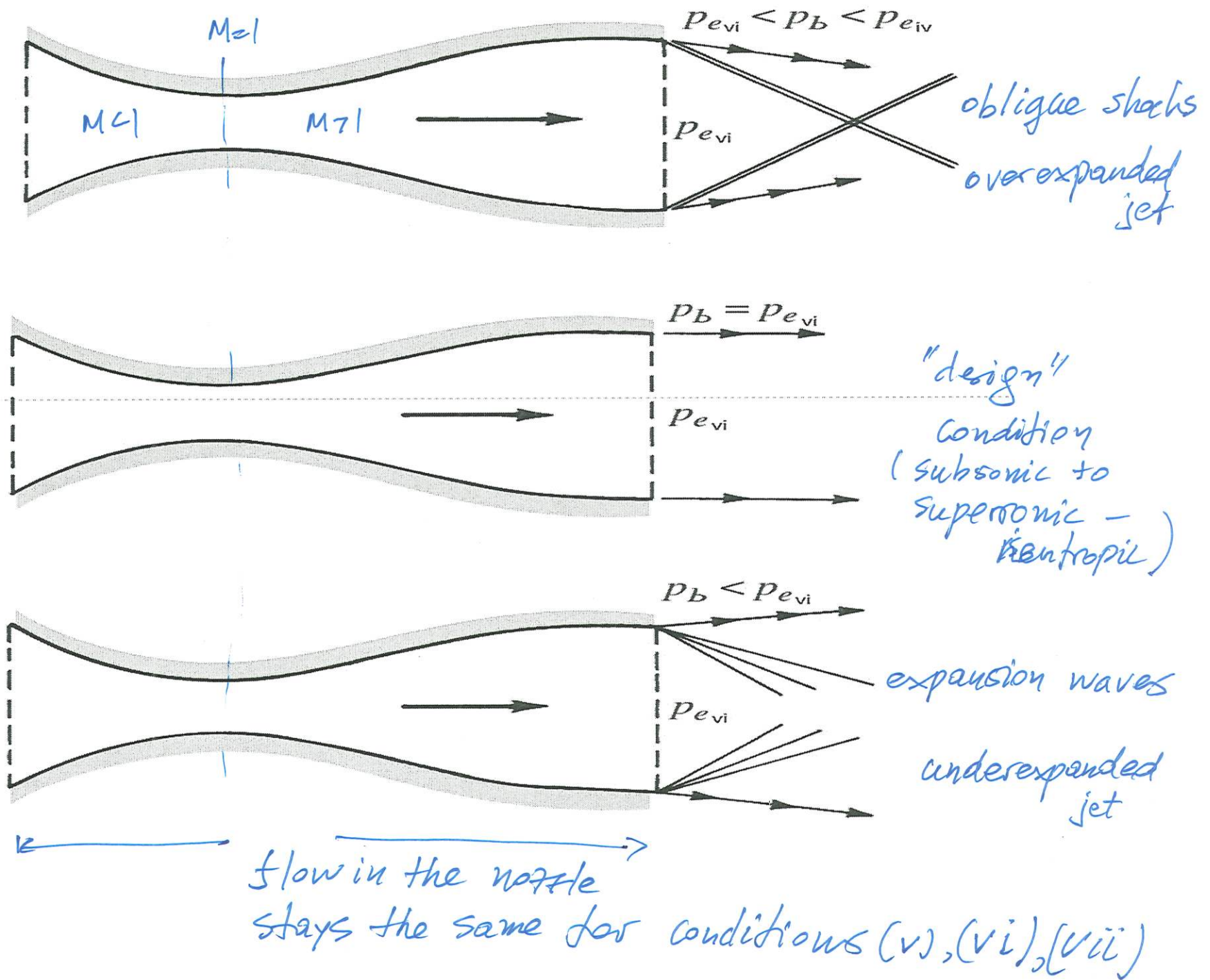
vi) if $p_b = p_{e_{vi}}$ normal isentropic exit (subsonic-supersonic)

vii) $p_b < p_{e_{vi}}$ - underexpanded jet

expansion waves after the nozzle (p_e can not go lower than $p_{e_{vi}}$)

note: $p_b \neq p_e$ for conditions (v) and (vii)

^
only



Problem 10.1: The reservoir pressure and temperature for a convergent-divergent nozzle are P_t and T_t 5 atm and 520° R, respectively. The flow is expanded isentropically to supersonic speed at the nozzle exit. If the exit-to-throat area is 2.193. Calculate the following properties at the exit: M_e , p_e , T_e , ρ_e , u_e , p_{te} , T_{te} .

$$\frac{A_e}{A^*} = 2.193 \xrightarrow[\text{tables}]{\text{isentropic}} M_e \approx 2.3 \xrightarrow{\quad} \frac{p_e}{p_{te}} = 0.08; \frac{T_e}{T_{te}} = 0.486$$

$$P_t = P_{te} = 5 \text{ atm}$$

(isentropic)

$$T_t = T_{te} = 520^\circ \text{R}$$

(adiabatic)

$$p_e = \frac{p_e}{p_{te}} \cdot p_{te} = (0.08) \times 5 = \underline{0.4 \text{ atm}}$$

$$T_e = \frac{T_e}{T_{te}} \cdot T_{te} = (0.486) \times 520 = \underline{252.7^\circ \text{R}}$$

$$\rho_e = \frac{p_e}{RT_e} = \frac{(0.4)(2116)}{(1716)(252.7)} = \underline{0.00195 \text{ slug/ft}^3}$$

$$u_e = M_e \sqrt{\gamma R T_e} = 2.3 \times \sqrt{(1.4)(1716)(252.7)} = \underline{1792 \text{ ft/sec}}$$

a_e 779.2 ft/sec

Nozzle Flow Analysis

For a given area ratio: $\frac{A_e}{A_{th}}$, P_t , and P_b (or P_t/P_b)
 [$P_e = P_b$ except conditions v, vii]

- Determine P_t/P_b for subsonic choked flow (ii) $P_b = P_e$

$(P_t/P_b)_{ii}$
 isentropic tables (using A_e/A^*) - subsonic solution

- Determine P_t/P_b for perfectly expanded flow (vi) $P_b = P_e$

$(P_t/P_b)_{vi}$
 isentropic tables (using A_e/A^*) - supersonic solution

- Determine P_t/P_b for shock at the exit (iv) $P_b = P_e$

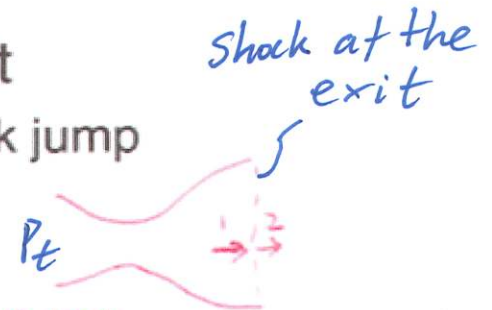
$(P_t/P_b)_{iv}$

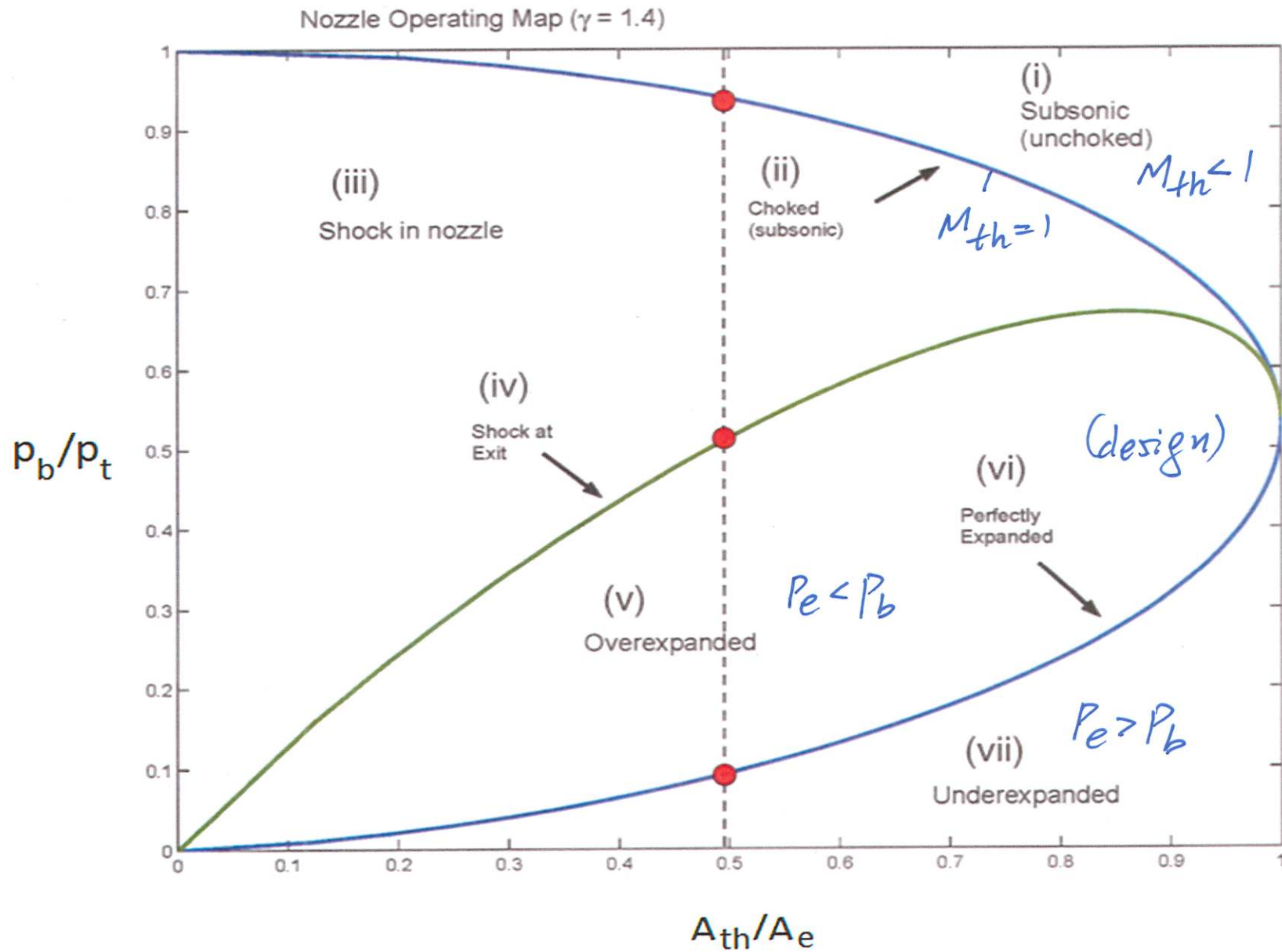
use M_{evi} for M_1 (Mach # before shock)
 normal shock tables: M_2 , $\frac{P_2}{P_1}$
 $\rightarrow (P_e)_{iv}$

Example (with #s)

For a given area ratio (e.g. $A_e/A_{th} = 2.0$):

- Determine P_t/P_b for (ii) subsonic choked flow
 - Isentropic subsonic solution with $A_{th} = A^*$
subsonic with $A/A^* = 2.0$ gives $P_t/P = \underline{1.0670}$
- Determine P_t/P_b for (vi) perfectly expanded flow
 - Isentropic supersonic solution with $A_{th} = A^*$
supersonic with $A/A^* = 2.0$ gives $P_t/P = \underline{10.6459}$ and $M_e = 2.1972$
- Determine P_t/P_b for (iv) shock at nozzle exit
 - Set M_1 equal to Mach number from (vi), use shock jump relations
 $M_1 = 2.1972$ gives $P_2/P_1 = 5.4656$
 - Then $P_t/P_b = (P_t/P_1) * (P_1/P_2) = \underline{10.6459} / 5.4656 = \underline{1.9478}$

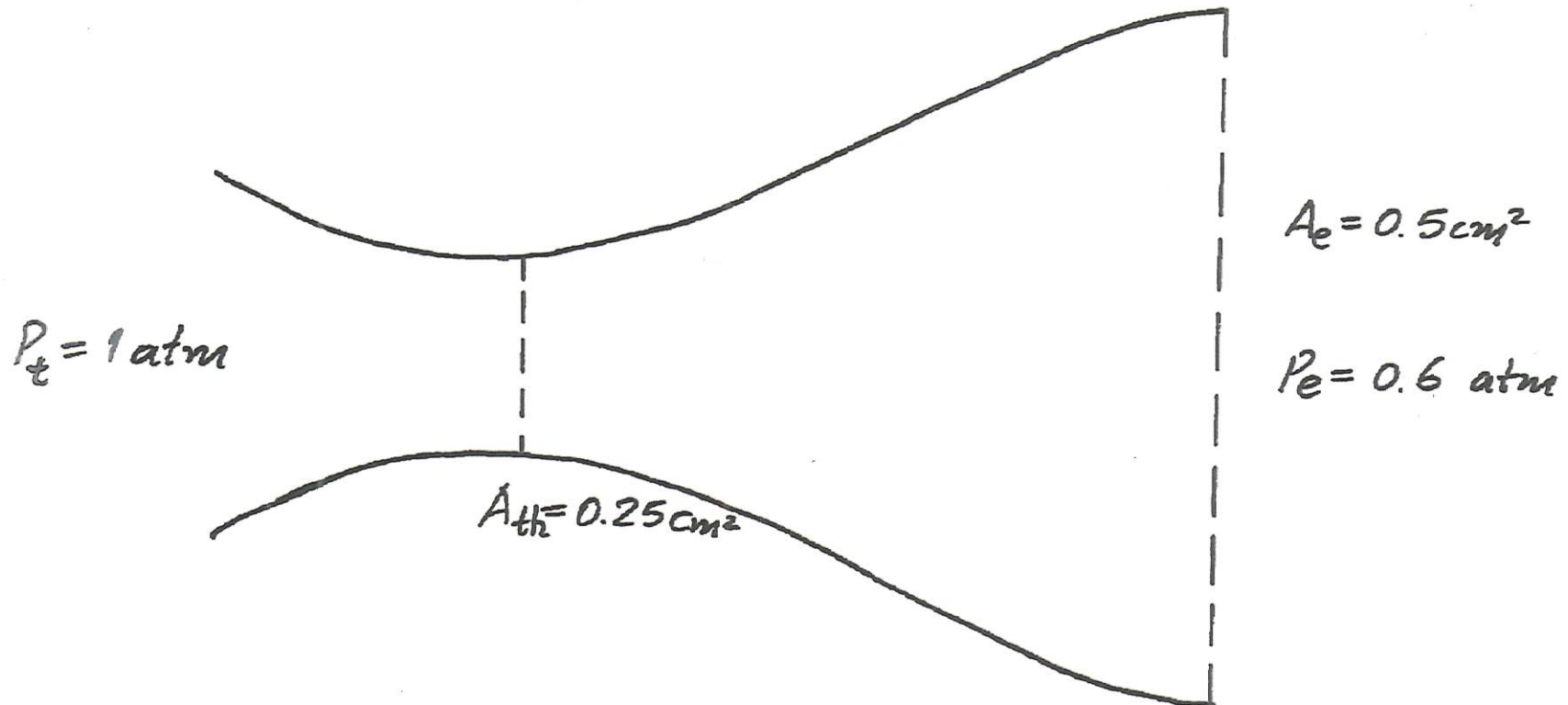




Example (finding normal shock location):

Consider a convergent-divergent duct with $A_e = 0.5 \text{ cm}^2$, $A_{th} = 0.25 \text{ cm}^2$

$p_t = 1 \text{ atm}$, $p_e = 0.6 \text{ atm}$; find M_e , shock position, strength



Isentropic tables for $\frac{A_e}{A_t} = 2$:

$$\text{Subsonic: } M=0.31 \rightarrow \frac{p_t}{p_e} = 1.069 \Rightarrow p_{e_{ii}} = 0.935 \text{ atm}$$

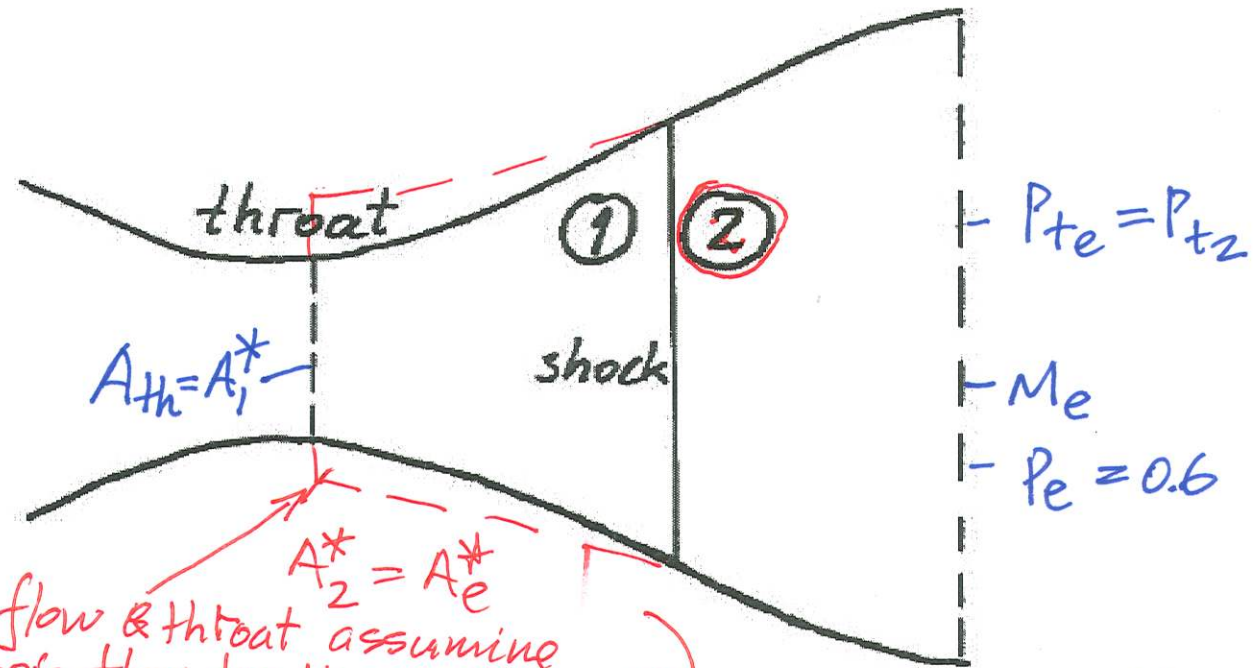
$$\text{Supersonic: } M=2.2 \rightarrow \frac{p_t}{p_e} = 10.69 \Rightarrow p_{e_{vi}} = 0.09 \text{ atm}$$

$$M_1 = 2.2 \Rightarrow (\text{shock tables}) \quad M_2 = 0.5471$$

$$\Rightarrow \frac{p_2}{p_1} = 5.480 \quad p_{e_{iv}} = 0.49 \text{ atm}$$

$\Rightarrow$ shock inside the nozzle, condition (iii)

$$\left(\begin{array}{l} p_{e_{ii}} = 0.935 \text{ atm} \\ p_{e_{iii}} = 0.6 \text{ atm} \\ p_{iv} = 0.49 \text{ atm} \end{array} \right)$$



(sonic flow @ throat assuming isentropic flow for the conditions (2))

because m is constant (8.45)

$p_t A^*$ is constant throughout the nozzle so

$$\Rightarrow \frac{p_{te} p_e}{p_e p_{t1}} = \frac{A_e A_{th}}{A_e^* A_e}$$

throat

exit

$$p_{t1} A_t = p_{te} A_e^*$$

$$\frac{A_1^*}{p_{t1}} \parallel \frac{A_2^*}{p_{t2}}$$

$$A_2^* > A_1^*$$

$$\Rightarrow \frac{p_e}{p_{t1}} \left(1 + \frac{\gamma-1}{2} M_e^2 \right)^{\frac{\gamma}{\gamma-1}} = \frac{A_{th}}{A_e} \frac{1}{M_e} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_e^2 \right) \right]^{\frac{(\gamma+1)}{2(\gamma-1)}}$$

$\swarrow p_{te}/p_e \text{ (8.36)}$
 $\searrow A_e/A_e^* \text{ (8.42)}$

(algebra)

$$\Rightarrow \frac{p_e A_e}{p_{t1} A_{th}} = \frac{1}{M_e} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \left(1 + \frac{\gamma-1}{2} M_e^2 \right)^{-\frac{1}{2}}$$

$\frac{1}{2} \left(\frac{\gamma+1}{\gamma-1} \right) - \frac{\gamma}{\gamma-1} = -\frac{1}{2}$

Cross-multiply and square both sides

$$M_e^2 \left(1 + \frac{\gamma-1}{2} M_e^2 \right) = \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \left(\frac{p_{t1} A_{th}}{p_e A_e} \right)^2$$

re-arrange as:

$$\left(M_e^2 \right)^2 + \frac{2}{\gamma-1} M_e^2 - \left(\frac{2}{\gamma-1} \right) \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} \left(\frac{p_{t1} A_{th}}{p_e A_e} \right)^2$$

Quadratic equation for M_e^2

- choose + sign to get $M_e^2 > 0$

$$M_e^2 = \frac{1}{(\gamma - 1)} \left\{ -1 + \left[1 + 2(\gamma - 1) \left(\frac{2}{\gamma + 1} \right)^{\left(\frac{\gamma + 1}{\gamma - 1} \right)} \left(\frac{p_{t1} A_{th}}{p_e A_e} \right)^2 \right]^{\frac{1}{2}} \right\}$$

eg. for M_e
 $\frac{p_{t1}}{p_e}, \frac{A_{th}}{A_e}$
 known

Using this formula for the current example: $M_e = 0.472$

isentropic tables: $\frac{p_{te}}{p_e} = 1.165$ $\frac{A_e}{A_e^*} = 1.398$
 $A_e^* = A_2^*$

$$\frac{p_{te}}{p_{t1}} = \frac{p_{te} p_e}{p_e p_{t1}} = (1.165)(0.6) = 0.699$$

for $\frac{p_{t2}}{p_{t1}} = 0.699$

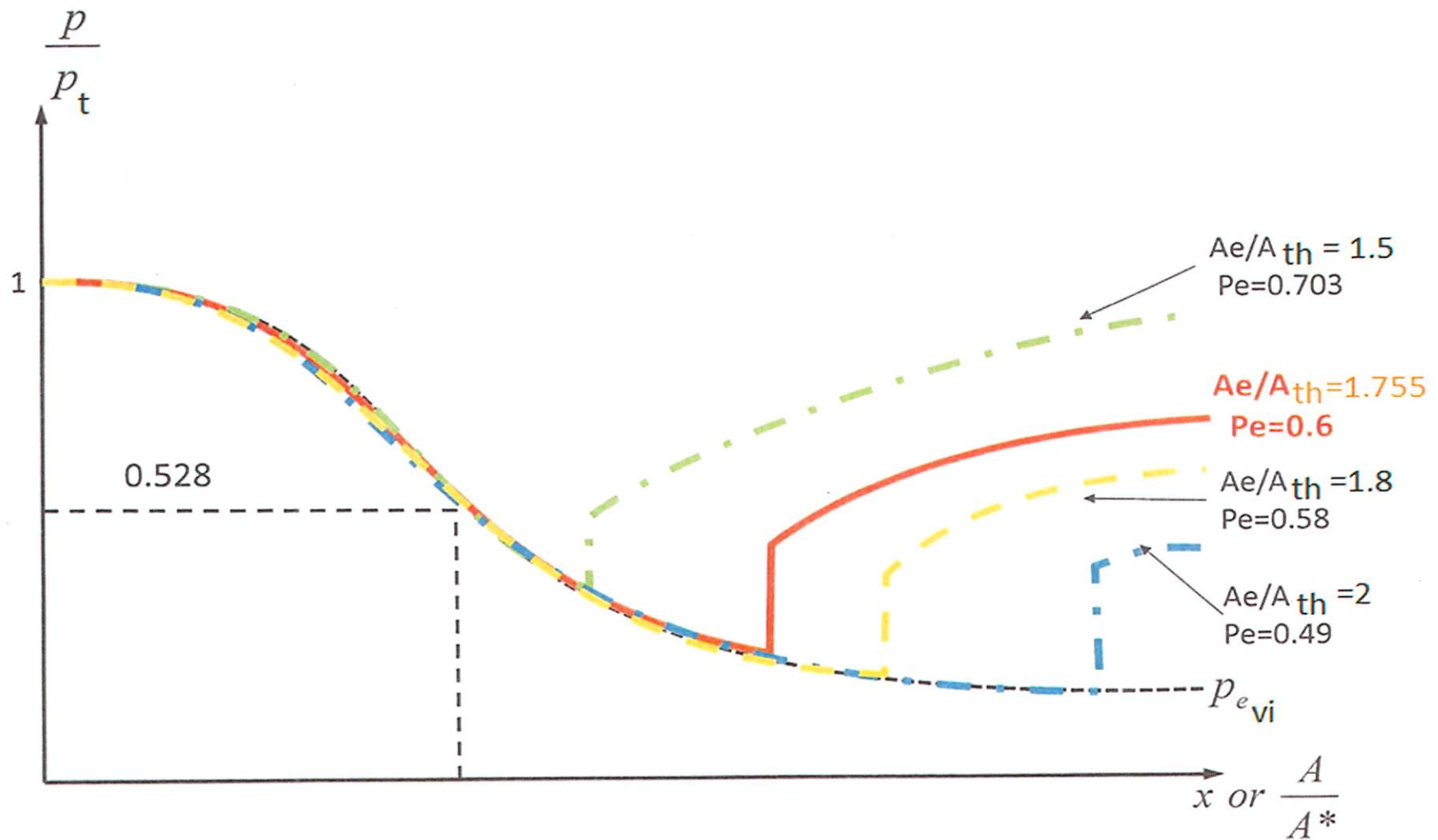
Normal shock tables: $M_1 = 2.047$ $M_2 = 0.5696$ $\frac{p_2}{p_1} = 4.721$

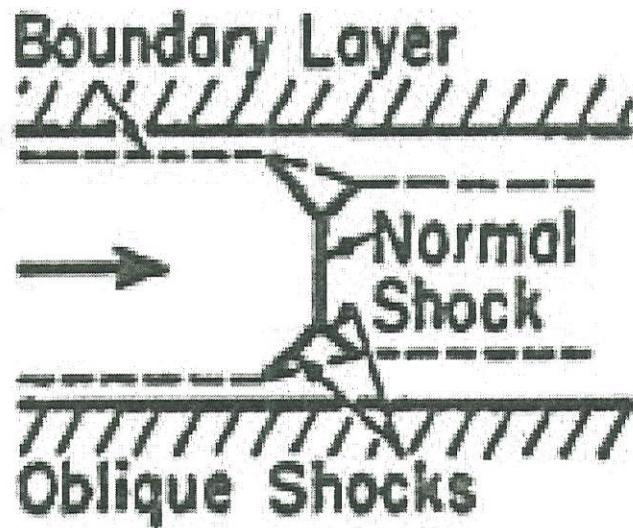
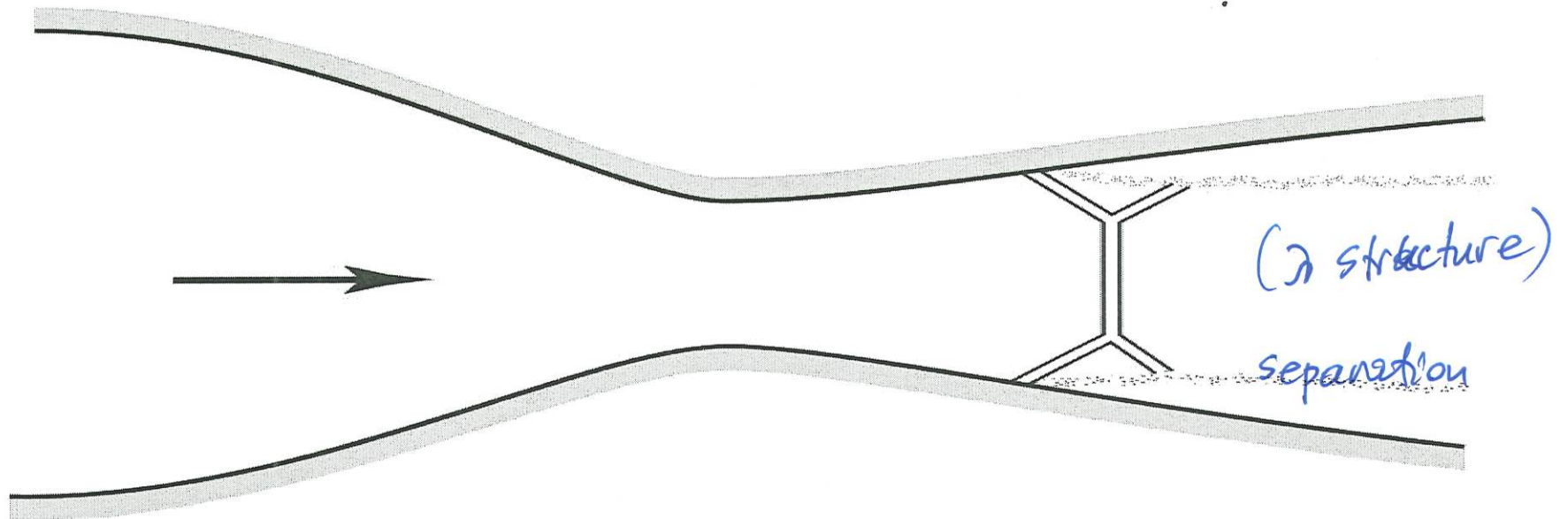
Isentropic tables for $M_2 = 0.5696 \rightarrow \frac{A_2}{A_2^*} = 1.227$

$$\frac{A_2}{A_{th}} = \frac{A_2}{A_2^*} \frac{A_2^*}{A_e} \frac{A_e}{A_{th}} = (1.227) \left(\frac{1}{1.398} \right) \left(\frac{0.5}{0.25} \right) = 1.755$$

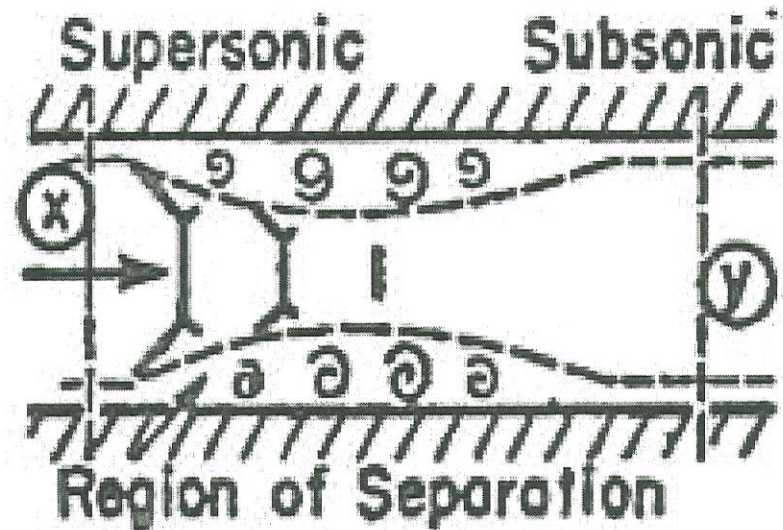
(shock location)
 (given)

shock locations ; stronger shocks near the exit



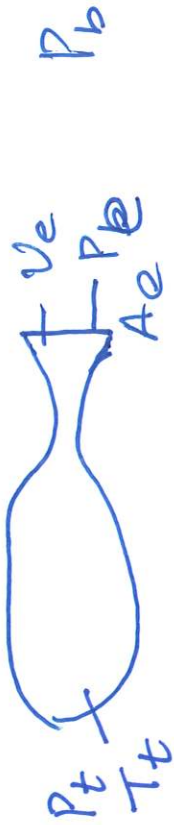


(Shapiro)



multiple shocks possible
when 1st shock is strong

Thrust:



Conservation of momentum:

$$\frac{\partial}{\partial t} \underbrace{\iiint_V \rho \mathbf{V} dv}_{\text{steady}} + \underbrace{\iint_S \rho \mathbf{V} (\mathbf{V} \times \mathbf{n}) dS}_{\text{entrance } V=0} = - \iint_S p \cdot \mathbf{n} dS + \iint_V \rho \mathbf{f} dv + \mathbf{F}_{\text{visc}} + \mathbf{F}_{\text{ext}}$$

neglect gravity

assume inviscid

$$\iint_S p \cdot \mathbf{n} dS = \iint_S (p - p_b) \cdot \mathbf{n} dS + \iint_S p_b \cdot \mathbf{n} dS \stackrel{=0}{=} (p_e - p_b) A_e \mathbf{e}_x$$

x-component

$$= \iiint_V (\nabla \cdot \mathbf{p}) dV \quad \text{gradient theorem}$$

$$\text{Thrust} = \rho U_e^2 A_e + (p_e - p_b) A_e = \dot{m} U_e + (p_e - p_b) A$$

$$C_T = \frac{\text{Thrust}}{p_t A^*}$$

Thrust coefficient:

remember:

$p_e = p_b$ except (v), (vii)